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Smart Voting Machine using Fingerprint Sensor and Face Recognition

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***Abstract* -Election is the mechanism which is used by people to choose a leader. The citizens can cast their vote either through ballet paper or Electronic Voting Machine [EVM]. Using either way of voting has a chance to misuse the other votes. As a result, this research work proposes a fingerprint sensor and face recognition-enabled electronic voting system. This helps to conduct a safe election without rigging as fingerprint and face are used as identification features. The vote caster fingerprints will be taken and it will be extracted and saved in the database for the voter registration and authentication processes. Storing the information in the database will prevent more than one registration from a single person. At the time of voting or during voting day, the individual person need to scan his/her fingerprint and it will be compared with the fingerprint database if the fingerprint is found in the fingerprint database. By using the Voter identity number and fingerprint, numerous duplicate registrations can be stopped and this will lead to high rate of actual success during voting. Voters can cast their ballots from any part if world by using their voter's unique identifier, authentication responses given on enrollment, and a token key supplied immediately for every member via the specially designed election web module. The administrator will send the token to the linked candidate's Email address. This proposed project has been implemented on University of Ibadan. Finally, the University of Ibadan had successfully completed the online voting system project by eliminating the occurrence of rigging. Some extra features are added to the project when compared with the existing system.**

Keywords-Face Recognition, Fingerprint Sensor, and Voting Machine.

I. INRODUCTION

India is the world's most populated democratic country, with democratic values based on the will of the people. Elections must be conducted without rigging in order for the country's future to be bright. Elections is one of the method existing from past few years to choose a leader based on their preferences with whom they wanted to be ruled. The existing voting technique employs the voter's ID or any other document as identity evidence to physically count the ballots. To decrease human costs and to increase security, the voting system is currently being converted to an electronic voting machine (EVM) as it has reusable nature which lowers election-related costs and facilitates large-scale election operations. The EVM system, on the other hand, has a number of security weaknesses and is more vulnerable than the old approach. More authentication, confidentiality, and voter privacy, among other things, which are required by the security system. Because of the verification process, experts had a lot of unfavorable things to say about e-voting. In the suggested model, a fingerprint sensor is devised and deployed to increase authentication and confidentiality. Every person has their own set of fingerprints. The fingerprints which we collected through the fingerprint sensor are stored in Arduino microcontroller. Votes are manually counted in a conventional approach. The count is also done automatically by the CPU in this system, preventing duplication or inaccurate counting. The voter need to enroll his fingers before the voting and it will be stored in database. Our suggested system's ultimate goal is to use fingerprints to authenticate the target person. People must register a few days before the voting procedure, which uses a fingerprint sensor, the key information about a person's authorization is enrolled by looking at their voter id. The data is collected and shared with the district/Taluk office via a server. On Election Day, the voter has the option of voting at any of the voting stations. The voter must verify their identity, andIf their fingerprint matches the data on the server, they will be able to cast their vote. The total tally of the votes at a particular station will be send to the district offices at the day end. The present method what we are using for conducting elections is the ballot paper. In this existing system we need to mark on the symbol of our leader and we need to through it in the ballot box. At the end of the day all the ballot boxes will be moved to one place where those boxes will be opened before counting. In the

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traditional voting system, if we are away from our polling station on polling day for any reason, we will be unable to vote. This was a key flaw in the traditional system that reduced the voting percentage. Electronic voting machines (EVMs) have recently been invented by scientists. Many drawbacks have been addressed by employing EVMs, such as the introduction of a ballot paper machine in place of the traditional ballot box, as well as a reduction in counting time. However, the option of using a voting system that could be accessed from anywhere which was not available earlier. The proposed approach allows voters to vote from any location. This increases the voting percentage. The primary goal of the proposed system is to provide voting flexibility, RFID tag identification, false card detection, data integrity, and avoid repetitive voting resulting in increase of total voting % to elect competent leaders to grow our country. In the current voting system, electronic voting machines (EVMs) are used. The voting unit, which is used by the voter, and the control unit, which is controlled by the control officer, make up the EVM. The two devices are connected by a five-meter wire.

On the voting unit, each candidate has a blue button, and each voting unit can accommodate up to 16 candidates. As a result, four units will be required to accommodate 64 candidates. Three buttons are located on the control unit's surface: one for releasing a single vote, one for viewing the total votes cast so far and one for stopping the voting process. The result key is concealed and sealed, and can only be pushed after pressing the close button. The aforesaid EVM system has some drawbacks, such as inconveniency, insecurity, human meddling, fraudulent votes, and low voter turnout, which our proposed method can overcome. In our proposed system, each station has its own voting unit, as well as a centralized unit. A person from any territory, regardless of origin, can vote at any of the stations. As a result, voters have more options because remote voting is allowed. In rare cases, the voting machine may malfunction, resulting in the loss of data. Our solution, on the other hand, uses data centralization, which means that after voting, we transfer all voting details to the proper server, preventing human interaction. To transfer voting information, we employ GSM technology. We can eliminate fraudulent voting and establish strong security with this approach, and we can easily determine who has voted and who has not. Elections are crucial in India because they determine whether a new administration is formed at the state or national level. Paper ballots and electronic voting machines are used in today's elections. However, there are two disadvantages to this electoral system. One possibility is that votes are duplicated or miscounted, resulting the system's poor performance. Another reason for India's low voting rate is that many voters are unable to vote because they are unable to reach their voting both. Even alternative remote voting is available in the system, the system stability is a major concern. As a result, a new fingerprint authentication voting, voters will able to vote more quickly and with greater confidence in the electoral process, as well as vote from anyplace in our country. This will significantly boost the number of people voting in our country. As a result, the finger print feature has been incorporated into the computerized e-voting system to improve security, secrecy, dependability, and accuracy.

II. LITERATURE SURVEY

A. Design and implementation of convenient and compulsory voting system using finger print sensor and GSM technologies

In this paper, authors have eliminated rigging of voting. They have attached the GSM module with the fingerprint sensor where the fingerprint's were collected and stored in Arduino for further use. Here they have used Arduino for a faster response at the time of election. In this paper, the main drawbacks of using is the GSM module which is very slow for communication and sometimes votes are missing. This is possible in poor networks.

B. Convenient and Secured Voting System

In this work, they have built a voting system that includes an RFID tag and an RFID reader for authentication, as well as a voting machine that includes a microcontroller and various input and output devices, as well as GSM technology. In this paper main drawbacks RFID key and GSM module. If RFID key is missed then voter can't cast their vote.

C. A Secured Biometric Voting System Using RFID Linked with the Aadhar Database

In this project, they proposed using an RFID tag instead of a voter ID. The RFID tag will be scanned and match the finger with the Aadhar database. Now the person will be allowed to scan their finger. If the finger matches with the Aadhar database, the person can cast their vote. The disadvantage of the project is that we have to create RFID tags for each individual person, which is costly.

D. Design and Implementation of Arduino Based Voting Machine

They will collect finger prints from the people and store them in a database. Then the voter will be provided with a unique ID. While people are casting their votes, they need to enter the unique ID and they need to scan their finger prints. Then it will compare the unique ID and fingerprint that are stored in the database and cast their vote. The disadvantage to this project is that there must be a voting officer and he needs to feed the details of a person to the machine. The person needs to scan their fingers two times in two places.

E. Electronic Voting System using Biometrics, Raspberry Pi and TFT module

In this project, the voter needs to enter his Aadhar Number on the display, which is connected to the Raspberry Pi. Then the voter needs to scan his fingers and his fingers are compared to a fingerprint library which is present in the database. The vote count is incremented and the vote is also encrypted. The user needs to enter his Aadhar details on the touch screen available for voting.

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For educated people, it is easy to operate, but for the uneducated, it's not that much easier to operate.

F. A Novel User Layer Cloud Security Model based on Chaotic Arnold Transformation using Fingerprint Biometric Traits

In this project, multitude of users access cloud by storing confidential information in the virtual storage platform or accessing vital data from archives, security and privacy is of prime concern. This has been taken as the motivation of this research work. An effective Chaotic based Biometric authentication scheme for user interaction layer of cloud is proposed and implemented
in this research paper.

G. Soft computing based autonomous low rate DDOS attack detection and security for cloud computing

The DDOS attacks are highly mature and continue to grow at a high speed making the detecting and the countermeasures a challenging task. So the paper uses the soft computing based autonomous detection for the Low DDOS attacks in the cloud architecture

III. EXISTING WORK

In this existing system, a secure voting machine is created using the AADHAR id. Here, the biometricsof a person will not be compared with the Aadhar database. In this system, face recognition is not implemented, so it would not be that secure.

IV. PROPOSED WORK

In our proposed project, a secure voting machine is created using the ADDHAR id. Here, the biometrics of a person and finger print will be compared with the Aadhar database. If the biometrics and finger print match without any error, the person is allowed to vote. Otherwise, the person will not be allowed to vote.

V. BLOCKDIAGRAM

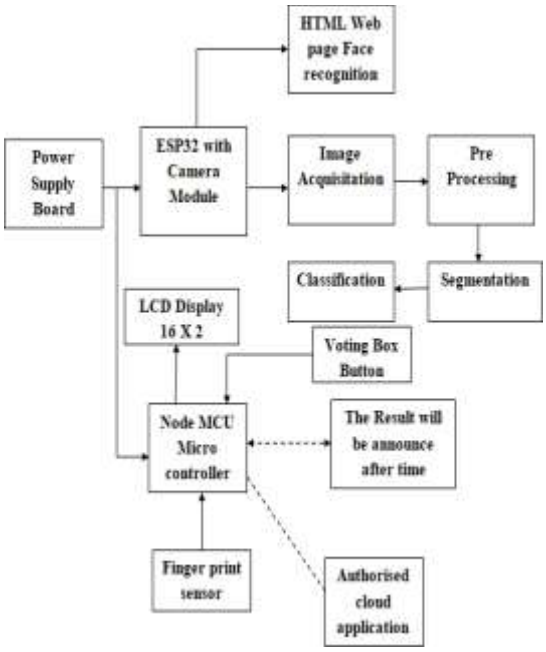

Figure 1: Block Diagram

VI. FLOW CHART

1. First the user will be checked with government identity proof
2. The voter's fingerprint and face recognition will be taken before election and stored in cloud
3. On the voting day the user needs to switch on the machine and scan his finger and face
4. The data stored in the cloud will be matching will the voter's fingerprint and face
5. If the data matches the voter can vote to his Favorite leader

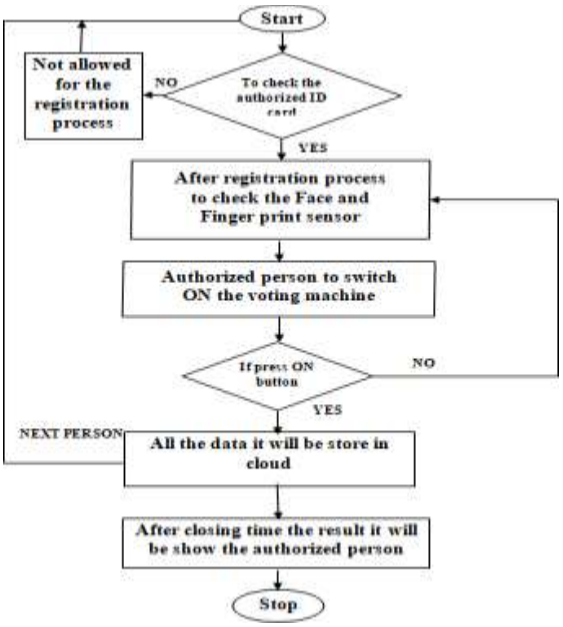

Figure 2: Flow chart

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VII.HARDWARE DESCRIPTION

A. ESP32 with camera module

Based on the ESP32 microcontroller, the ESP32-CAM is a small, low-power camera module. It comes with an OV2640 camera and a TF card slot. The ESP32-CAM is suited for Internet of Things (IoT) applications such as wireless video surveillance, WIFI picture upload, QR identification, and other intelligent IoT applications.

The ESP32-CAM is a low-cost, tiny ESP32-based development board with an integrated camera. It's ideal for Internet of Things (IoT) projects, prototypes, and do-it-yourself projects. The board integrates Wi-Fi, conventional Bluetooth, and low-power BLE with two high-performance 32-bit LX6 CPUs. It has a 7-stage pipeline architecture, an on-chip sensor, a Hall sensor, a temperature sensor, and other features, as well as an 80 MHz to 240 MHz main frequency adjustment range. It can be used as a master to build a standalone network controller or as a slave to add networking capability to existing devices using other host MCUs.Power consumption of esp32 camera module is 5V. It complies with Wi-Fi 802.11b/g/n/e/i and Bluetooth 4.2 specifications. The ESP32-CAM is suitable for a wide range of IoT applications. **Here we have used it for capturing the face id of individual which is stored in the cloud storage using NODE MCU MICROCONTROLLER**. It's suitable for smart home devices, industrial wireless control, wireless monitoring, QR wireless identification, wireless positioning system signals, and other IoT applications. It's a fantastic option for IoT applications.

Figure 3: Camera Module

B. JUMPER WIRES

Jumper wires are wires with connector pins on both ends that can be used to link two locations without using solder. With breadboards and other prototype tools, jumper wires are widely utilized to allow for fast circuit adjustments as needed. It's a simple procedure. The most basic of all electrical components are jumper wires. Jumper wires are wires with connector pins on both ends that can be used to link two locations without using solder. With breadboards and other prototype tools, jumper wires are widely utilized to allow for fast circuit adjustments as needed. It's a simple procedure. The most basic of all electrical components are jumper wires.

Figure4: Jumper Wires

C. NODE MCU MICROCONTROLLER

MCU stands for microcomputer and is made up of an integrated circuit chip composed of Metal oxide Semiconductor (MOS). It is made up of a microprocessor and peripherals for input and output. It's commonly used in embedded software and other general-purpose software. Microcontrollers can be incorporated into any automation and control system. The power consumption of NODE MCU MICROCONTROLLER is 5V ,maximum voltage dropout is 1.1V @ 800mA and typical Quiescent current is 5Ma. The Node MCU is an open-source firmware and development kit for prototyping and building Internet-of-Things (IoT) devices. It includes software for the ESP8266 Wi-Fi SoC from Express if Systems, as well as hardware for the ESP-12 module. The firmware uses the Lua scripting language. It is built on the e Lua project and developed using the Express if Non-OS SDK for ESP8266.

Figure 5: Node MCU Microcontroller

D. LCD Display

A liquid crystal display (LCD) is an electrical display module that uses a liquid crystal to produce a visible image. The 162 LCD display is a simple module that can be found in a wide range of DIY projects and circuits.

A two-line display with 16 characters per line is represented by the number 162. On this LCD, each character is displayed in a 5x7 pixel matrix.

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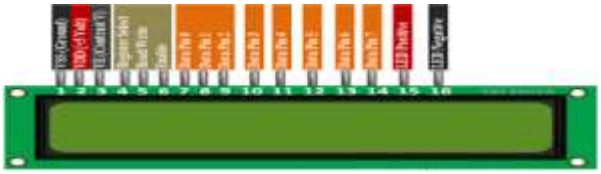

Figure 6: LCD Display

E. BUTTON

A pushbutton is a component that, when pressed, connects two points in a circuit. When you press the button, the example turns on an LED. Three cables are connected to the Arduino board. The first connects one leg of the pushbutton to the 5-volt supply via a pull-up resistor (here 2.2kohms). The second connects the pushbutton's matching leg to the ground. The third wire connects to a digital i/o pin (in this case pin 7) that reads the state of the button. Because there is no connection between the two legs of the pushbutton when it is open (unpressed), the pin is linked to 5 volts (via the pull-up resistor) and we read a HIGH. When the button is pressed, a connection is made between its two legs, connecting the pin to ground, resulting in a LOW reading. (The pin is still linked to 5 volts, but because of the resistor in between, it is "closer" to ground.) Attach three wires to the circuit board. The first two, red and black, link to the 5-volt supply and ground through the two long vertical rows on the breadboard's side. A third wire connects digital pin 2 to one of the pushbutton's legs. A pull-down resistor (here 10K ohm) links the corresponding leg of the button to the ground. The 5-volt supply is connected to the other leg of the button.

Figure 7: Button

F. FINGER PRINT MODULE

The Fingerprint Module is a serial fingerprint scanner that connects to the PC's com port directly. The MAX232IC allows the R305 Fingerprint Sensor to be simply connected to any controller. This fingerprint scanner is capable of storing and comparing fingerprints and providing the desired output as a result. This Arduino fingerprint sensor project can be used for a variety of applications including door security, forensics, crime investigation, personal identity, attendance systems, and more. The entire system is based on a basic technique known as the matching algorithm, which compares previously recorded fingerprint templates against users' fingerprints for authentication purposes.Biometrics will aid in project security, and this all-in-one optical fingerprint sensor will make fingerprint recognition and verification a breeze. In these modules, which are commonly found in safes, a high-powered DSP processor handles image rendering, calculating, feature-finding, and searching.Power consumption depends on the voltage level used for the sensor, the current drawn by the device, the I/O mode used and clock rate selected. For SPI mode running at 2.7 volts and 12 MHz (700+ frames/s, or a 6-inch/s finger sweep rate), the system will consume 32 mW during the active finger sweep.Send data packets to any microcontroller or system via TTL serial to snap photographs, identify prints, hash, and search. The onboard FLASH memory can store up to 162 finger prints, so you may enroll new fingers right away. The lens has a red LED that illuminates during a photo to let you know it's operating. It also comes with easy-to-use Windows software that makes testing the module a breeze; you can even enroll and view a fingerprint picture on your computer screen using the software. But, of course, we wouldn't just leave you with a datasheet and a "best of luck!" — we built a whole Arduino library that you can use in under10 minutes. The library has the ability to enroll and search, making it ideal for any project. We've also published a comprehensive guide to wiring and use.

Figure 8: Finger Print Module

VIII. ACCURACY MEASURES OF PROPOSED WORK

Accuracy Measure of proposed work is 95 Percentage

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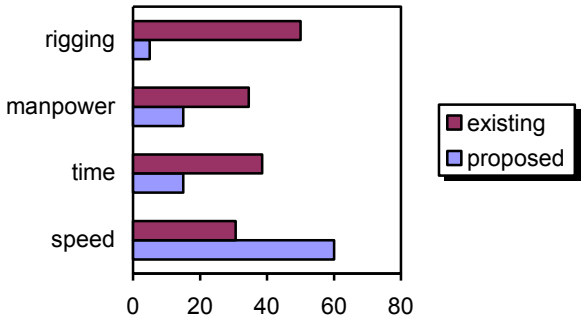

Figure 9:Comparison between existing method and propoased work

IX. COMPARISON BETWEEN PROPOSED SYSTEM AND EXISTING SYSTEM.

TABLE 1.

S.NO	COMPARISON BETWEEN PROPOSED SYSTEM AND EXISTING SYSTEM		
	Parameter	***Proposed***	***Existing***
1	Man power	Less	More
2	Speed	More	Less
3	Time to declare election results	Less	More
4	Rigging of Votes	Not Possible	Possible

Comparison between proposed system and existing system.

X. RESULTS AND CONCLUSION

This research paper discusses fingerprint-based authentication for secure voting. For a more effective and transparent election process, finger print-based authentication and face recognition-based data exchange are used to address security concerns in electronic voting machines. The proposed strategy achieves better results and gives candidates more opportunities to vote in any place. It reduces human error and difficulties throughout the voting count procedure. The information must be encrypted and saved to increase the system's security.This helps to prevent data leaks while also providing better privacy to users. The votes are collected and will transfer to the head office of the voting stations in a secured manner. This avoids votes leakage. This proposed system eliminates the possibility of bogus voting. The fingerprint acquisition method is where this prototype's limitation lies. Because of age, job, and other variables such as skin disorders, the finger print is tampered with.

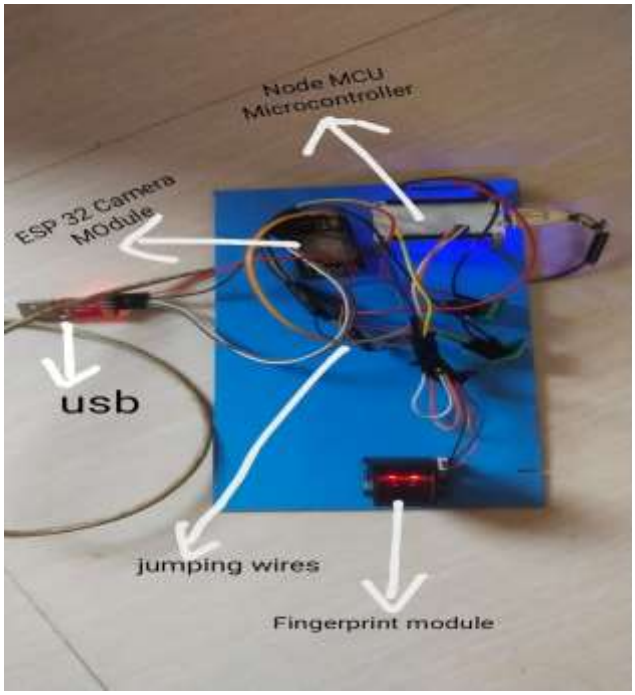

Figure 10: Hard Ware Kit

Figure 10.1: A) **Voters Welcome Portal**

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Figure 10.1: B) **Voters Welcome Portal**

Figure 10.1: C) **Voters Welcome Portal**

Figure 10.1:D)**Voters Welcome Portal**

XI. FUTURE WORK

The Research could expand in the future to include the use of an iris scanner in conjunction with a fingerprint scanner for more safe and reliable operation. It has the potential to alleviate the limits of finger print-based operations. In future we are going to propose an e voting system which is having no involvement of the human at the time voting to misuse the votes of the people. The correlation between fingerprints can be obtained in order to minimize server storage space and improve system performance. The designed unit is put to the test. We employed a face recognition technique for biometric features; however, any other biometric technique might be used. We tested the performance on 100 distinct faces and found it to be great. This could be a good way to adapt high- security voting equipment. In this paper, we have proposed a safe and secure voting system that is superior to the previous system. This technology will prevent illegitimate voters from voting and will provide faster and more accurate results. It also verifies the voter's eligibility and prohibits the same person from voting several times. It enables a person to vote from any location. We currently use finger prints, but we may be able to use iris technology in the future.

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